

<!-- Source: docs\manual\00-cover-and-document-control.md -->

Gathetr Owner's Technical Operations, Development, Maintenance and Repair Manual

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Firebase project: gathervibeshub

Production URL: <https://gathervibeshub.web.app>

Document Control

This manual covers the code and configuration present at the recorded commit only. It does not claim to describe later commits unless regenerated.

Source documents in `docs/` are authoritative. The PDF is a generated readable snapshot.

Protected Boundaries

- Do not expose `.env.local`, cookies, tokens, service account JSON, or private Firebase keys.
- Do not use CPB for synthetic QA or fake data.
- QR payload format must remain `GSV:TICKET:{ticketCode}` unless explicitly approved and migrated.
- Firestore Rules, Auth, Hosting, Functions, and production data require explicit validation and approval before deployment.
- This Phase 3 documentation generation requires no Firebase deployment and makes no runtime behavior change.

<!-- Source: docs\manual\01-quick-start-and-emergency-reference.md -->

Quick Start and Emergency Reference

Immediate Symptom Table

Symptom	First check	Next manual section
---	---	---
Blank page	Browser Console and Network; check stale chunk or Firebase config	Troubleshooting: blank page or dynamic import failure
Permission denied	Signed-in account, `settings/accessControl`, staff assignment, Firestore rule line	Permissions and Security; Permission runbooks
Login fails	Firebase Auth provider, authorized domain, popup/redirect state	Firebase Authentication; Login failure runbook
Scanner fails	Camera permission, HTTPS, QR payload, assigned event, ticket code lookup	Imports/Exports/QR; Scanner runbook
Build fails	Terminal error and changed files	Build Deployment and Recovery
Production fails	Hosting deployment, asset cache, Firebase project, Console	Emergency recovery runbooks

Safe First Commands

```
git status --short --branch
git rev-list --left-right --count main...origin/main
npm run lint
npm test
npm run build
```

Do not run deployment commands while diagnosing unless the fix is known, tested, and approved.

<!-- Source: docs\manual\02-application-overview.md -->

Application Overview

Gathetr is a private React/Firebase event operations application for planning, guest management, registration payments, tickets, check-in, operations, run of show, resources, documents, contacts, reports, imports, message preparation, Settings, and System QA.

The active synthetic training event is `CODEX\_DEMO - Full System Walkthrough`. CPB is real historical production data and must not receive synthetic writes.

Main User Groups

- Protected Owner: UID-protected permanent owner access.
- Approved Organizer: active approved email in `settings/accessControl`.
- Staff roles: active `staffProfiles/{uid}` plus active event assignment.
- Scanner: assigned-event check-in-only route and narrow Firestore update path.

Stack

```
| Layer | Actual technology |
| --- | --- |
| Frontend | React ^19.2.0, Vite ^7.2.4, React Router ^8.3.0 |
| Styling | Tailwind CSS ^4.1.17 through @tailwindcss/vite |
| Backend services | Firebase client SDK ^12.6.0 |
| Database | Cloud Firestore with local emulator tests |
| Authentication | Firebase Authentication with Google and email/password |
| Hosting | Firebase Hosting classic static SPA rewrites to index.html |
| QR | qrcode ^1.5.4; scanner uses html5-qrcode ^2.3.8 |
| Excel parsing | read-excel-file ^9.2.0; xlsx package intentionally absent |
```

<!-- Source: docs\manual\03-system-architecture.md -->

System Architecture

High-Level Flow

flowchart TD

```
User[Owner, organizer, staff, scanner] --> Browser[Browser / PWA shell]
Browser --> React[React 19 + Vite app]
React --> RouteGate[ProtectedRoute and canViewRoute]
RouteGate --> Auth[Firebase Authentication]
Auth --> Access[Access resolver: settings/accessControl + staff profile + assignment]
Access --> FirestoreRules[Firestore Security Rules]
FirestoreRules --> Firestore[(Cloud Firestore)]
React --> Hosting[Firebase Hosting static assets]
React --> QA[Local emulator tests and Playwright QA]
```

Important Boundaries

- Frontend route checks improve user experience but do not replace Firestore Rules.
- Firestore Rules are the backend security boundary.
- Audit logs are append-only evidence and are coupled to business mutations.
- Message Builder is copy-only; it does not send email.
- Documents are references/links and metadata only; no Firebase Storage upload is active.

<!-- Source: docs\manual\04-project-file-map.md -->

Project File Map

Feature Ownership Map

Feature	Main files	Related tests	Risk
---	---	---	---
Authentication and access resolution	src/lib/firebase.js; src/auth/AuthProvider.jsx; src/auth/ProtectedRoute.jsx; src/utils/accessRoles.js; src/config/protectedOwner.js	tests/auth-reliability.test.js; tests/protected-owner-authorization-matrix.test.js; tests/settings-access-management.test.js	High
Events and working event	src/pages/EventsPage.jsx; src/services/eventService.js; src/events/ActiveEventProvider.jsx; src/utils/eventPlanning.js	tests/event-utils.test.js; tests/phase26-event-categories-capabilities.test.js	High
Tasks and deadlines	src/pages/TasksPage.jsx; src/services/taskService.js; src/utils/taskWorkflow.js	tests/phase2-guided-event-setup-task-deadline.test.js; tests/task-workflow-rules.test.js	Medium
Guests and registrations	src/pages/RegistrationsPage.jsx; src/services/registrationService.js; src/utils/validators.js; src/utils/registrationMetrics.js	tests/registration-utils.test.js; tests/registration-metrics.test.js	High
Registration payments	src/pages/PaymentsPage.jsx; src/utils/financeUtils.js; src/utils/paymentStatus.js	tests/phase23b-payments-operations-boundaries.test.js; tests/payments-operations-refinement-2026-08.test.js	High
Review and reconciliation	src/pages/PaymentReconciliationPage.jsx; src/services/reconciliationReadService.js; src/utils/paymentReconciliation.js; src/utils/reconciliationWorkbook.js	tests/phase23c-payment-reconciliation.test.js	High
Tickets and QR	src/pages/TicketsPage.jsx; src/components/tickets/TicketQrCode.jsx; src/services/ticketService.js; src/utils/ticketUtils.js; src/utils/qrTicketUtils.js	tests/phase45-ticketing.test.js; tests/phase7-qr-checkin.test.js	High
Check-in and scanner	src/pages/CheckInPage.jsx; src/pages/ScannerPage.jsx; src/components/checkin/QrScannerPanel.jsx; src/utils/checkInUtils.js	tests/phase14-camera-checkin.test.js; tests/firestore-checkin-rules.test.js; e2e/workflows.spec.js	High
Imports and XLSX parsing	src/pages/ImportsPage.jsx; src/services/importService.js; src/utils/importUtils.js; src/utils/xlsxImport.js	tests/import-center.test.js; tests/import-center-workflow-upgrade.test.js	High
Operations ledger and commitments	src/pages/OperationsPage.jsx; src/services/operationsLedgerService.js; src/components/operations/PartnerCommitmentsPanel.jsx	tests/phase19-operations-productivity.test.js; tests/phase23b-payments-operations-boundaries.test.js	High
Run of Show	src/pages/RunOfShowPage.jsx; src/services/runOfShowService.js; src/utils/runOfShow.js	tests/run-of-show-resources-foundation.test.js; tests/run-of-show-resource-rules.test.js	Medium
Equipment and supplies	src/pages/ResourcesPage.jsx; src/services/eventResourceService.js; src/utils/eventResources.js	tests/run-of-show-resources-foundation.test.js; tests/run-of-show-resource-rules.test.js	Medium
Documents	src/pages/DocumentsPage.jsx; src/services/documentService.js; src/utils/documentRegistry.js	tests/document-contact-foundation.test.js; tests/document-contact-rules.test.js	Medium
Contacts and organizations	src/pages/ContactsPage.jsx; src/services/contactService.js; src/utils/contactDirectory.js	tests/document-contact-foundation.test.js; tests/document-contact-rules.test.js	Medium
Settings, staff, integrations	src/pages/SettingsPage.jsx; src/services/accessManagementService.js; src/services/staffManagementService.js; src/services/integrationSettingsService.js	tests/settings-access-management.test.js	High
Message Builder	src/pages/CommunicationsPage.jsx; src/utils/communicationsUtils.js	tests/phase6-communications.test.js	Medium
System QA	src/pages/QaPage.jsx; src/utils/qaHelper.js; src/utils/runtimeHealth.js	tests/production-qa.test.js; tests/settings-systemqa-tutorial-final-refinement.test.js	Medium
Tutorial/onboarding	src/tutorial/*	tests/onboarding-flow.test.js; tests/phase26-interactive-product-tour.test.js; e2e/tutorial.spec.js	Medium

Root Configuration

Path	Purpose	Risk
---	---	---
package.json	Scripts and dependencies for development, QA, E2E, Firebase deploy helpers, docs generation.	High
firebase.json	Auth provider notes, emulator ports, Firestore file references, Hosting headers and rewrites.	High
firestore.rules	Backend authorization and schema validation.	Critical
firestore.indexes.json	Firestore composite index configuration.	High
vite.config.js	Vite/React/Tailwind build configuration.	Medium
playwright.config.js	E2E browser test configuration and emulator env.	Medium
AI_AGENT_RULES.md	Permanent coding assistant governance.	High

<!-- Source: docs\manual\05-frontend-react-and-tailwind.md -->

Frontend React and Tailwind

Entry point: ``src/main.jsx``

Top-level app: ``src/App.jsx``

Authenticated shell: ``src/layout/AppShell.jsx``

Global styles: ``src/styles.css``

React Patterns

- Lazy-loaded route pages are declared in ``src/App.jsx``.
- ``ProtectedRoute`` blocks unauthenticated or unauthorized users.
- ``AppShell`` owns desktop sidebar, mobile drawer, mobile tab bar, page titles, Admin Search, page guidance, and `TutorialProvider` wrapping.
- Firebase writes are kept in service modules under ``src/services/``.
- Shared validation and display helpers live in ``src/utils/``.

Tailwind Usage

Tailwind is used directly through class names and the Vite Tailwind plugin. The app uses compact cards, dense grids, responsive ``sm``, ``md``, ``lg``, ``xl`` breakpoints, and semantic color usage tied to the Gather & Savor visual system.

Troubleshooting:

- Element not visible: check responsive utility prefixes and route/access gating.
- Mobile overflow: inspect fixed widths, tables, and long event names.
- Dynamic class not applying: avoid constructing class strings that Tailwind cannot see statically.
- Form accessibility issue: inspect associated label, id, name, and focus behavior.

<!-- Source: docs\manual\06-routing-and-navigation.md -->

Routing and Navigation

Routes are declared in `src/App.jsx`; page titles and navigation groups are in `src/layout/AppShell.jsx`.

Route Map

Route	Title	Purpose	Component/Gate	Access	Source
---	---	---	---	---	---
/dashboard	Event Overview	Current event status, priorities, and next actions	<DashboardPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/events	Events	Plan and organize every gathering	<EventsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/tasks	Tasks & Deadlines	Event-scoped work, blockers, and follow-up dates	<TasksPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/registrations	Guests & Registrations	Manage registration records and guest counts	<RegistrationsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/payments	Registration Payments	Review registration charges, payments, balances, and follow-up	<AssignedEventGate purpose="Payments"><PaymentsPage /></AssignedEventGate>	Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/payments/reconciliation	Review & Reconcile Records	Read-only workbook comparison before any correction	<PaymentReconciliationPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/tickets	Tickets	Assign ticket codes and prepare QR access	<TicketsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/check-in	Check-In	Track event-day attendance	<AssignedEventGate purpose="Check-In"><CheckInPage /></AssignedEventGate>	Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/operations	Operations	Track event-level money and obligations	<AssignedEventGate purpose="Operations"><OperationsPage /></AssignedEventGate>	Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/run-of-show	Run of Show	Event-day sequence, supplier arrivals, dependencies, and Now/Next	<AssignedEventGate purpose="Run of Show"><RunOfShowPage /></AssignedEventGate>	Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/resources	Equipment & Supplies	Equipment, supplies, packing, pickup, and return tracking	<AssignedEventGate purpose="Resources"><ResourcesPage /></AssignedEventGate>	Authenticated + assigned Working Event gate	src/App.jsx; src/layout/AppShell.jsx
/documents	Documents	Event document references, links, and evidence	<DocumentsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/contacts	Contacts & Organizations	Reusable people, businesses, and event relationships	<ContactsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/event-review	Reports	Read-only follow-up, payments, operations, and summary	<EventReviewPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/imports	Import Center	Bring in CSV exports and pasted table rows safely	<ImportsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/qa	System QA	System health, data checks, and safe test guidance	<QaPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/communications	Message Builder	Create, personalize, and copy event messages	<CommunicationsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx
/settings	Settings	Practical workspace and event defaults	<SettingsPage />	Authenticated route gate	src/App.jsx; src/layout/AppShell.jsx

```
flowchart LR
  Login[/login/] --> Protected[ProtectedRoute]
  Protected --> Shell[AppShell]
  Shell --> Home[/dashboard/]
  Shell --> Plan[/events /tasks /contacts /documents /run-of-show /resources/]
  Shell --> Guests[/registrations /payments /tickets /check-in/]
  Shell --> Ops[/operations /event-review /payments/reconciliation /imports /communications/]
  Shell --> Admin[/settings /qa/]
  Protected --> Scanner[/scanner assigned-event gate/]
```

Public route: `/login`.

Redirect aliases: `/security -> /settings`, `/reconciliation -> /payments/reconciliation`, `/reports -> /event-review`.

<!-- Source: docs\manual\07-firebase-authentication.md -->

Firestore Authentication

Firestore initialization is in `src/lib/firebase.js`. Authentication lifecycle is in `src/auth/AuthProvider.jsx`.

Auth Sequence

```
sequenceDiagram
    participant B as Browser
    participant A as AuthProvider
    participant F as Firebase Auth
    participant D as Firestore
    participant R as ProtectedRoute
    B->>A: App opens
    A->>F: setPersistence + onAuthStateChanged
    F-->>A: current user or null
    A->>D: read settings/accessControl
    A->>D: fallback read staffProfiles and assignments
    A->>A: resolve access level and default route
    A-->>R: user, loading, isAuthorized, access
    R-->>B: render page or redirect
```

Definitions:

- Authentication state: Firestore's current determination of whether a user is signed in.
- Authorization state: Gathetr's determination that the signed-in user has owner, approved organizer, or assigned staff access.

Flow

1. App initializes Firestore from Vite environment variables.
2. Auth persistence is set to browser local persistence.
3. Firestore reports the current user through `onAuthStateChanged`.
4. Gathetr reads `settings/accessControl` for approved organizer access.
5. If organizer access is not found, it reads `staffProfiles/{uid}` and active assignment docs for supported event IDs.
6. `getUserAccessLevel` builds the access object used by route gates and labels.
7. Firestore Rules still enforce every backend read/write.

<!-- Source: docs\manual\08-firestore-data-model.md -->

Firestore Data Model

This dictionary is based on current services, Firestore Rules, and tests. It does not include real production personal data.

Collections and Subcollections

Collection path	Purpose	Code that uses it	Rules reference	Security note
---	---	---	---	---
settings/accessControl	Authoritative approved organizer source plus metadata records.	src/auth/AuthProvider.jsx; src/services/accessManagementService.js; src/pages/SettingsPage.jsx	firestore.rules match /settings/accessControl	Protected Owner only for mutations; approved admins read.
settings/accessControl/history/{historyId}	Append-only history for organizer access changes.	src/services/accessManagementService.js	firestore.rules match /settings/accessControl/history/{historyId}	Protected Owner create; approved admins read.
settings/integrations	Supported integration status settings surfaced in Settings.	src/services/integrationSettingsService.js	firestore.rules match /settings/integrations	Protected Owner create/update; approved admins read.
events/{eventId}	Event records, planning data, capability configuration, and working-event source.	src/services/eventService.js; src/events/ActiveEventProvider.jsx	firestore.rules match /events/{eventId}	Approved admins manage; assigned staff can read assigned events.
events/{eventId}/staffAssignments/{uid}	Event-scoped staff assignment records.	src/services/staffManagementService.js; src/auth/AuthProvider.jsx	firestore.rules match /events/{eventId}/staffAssignments/{uid}	Approved admins manage; assigned user can read own active assignment.
events/{eventId}/tasks/{taskId}	Event-scoped tasks and deadlines.	src/services/taskService.js	firestore.rules match /events/{eventId}/tasks/{taskId}	Approved admins and assigned event managers manage; viewers can read.
registrations/{registrationId}	Guest registration, ticket, payment, and check-in status records.	src/services/registrationService.js; src/services/ticketService.js	firestore.rules match /registrations/{registrationId}	Approved admins manage; assigned scanners can perform narrow check-in updates.
auditLogs/{logId}	Append-only audit evidence for business mutations.	src/services/auditService.js	service write batches firestore.rules match /auditLogs/{logId}	Create only when matching a permitted target mutation; never update/delete.
operationsLedger/{ledgerEntryId}	Event-level Operations ledger, commitments, partners, in-kind support.	src/services/operationsLedgerService.js	firestore.rules match /operationsLedger/{ledgerEntryId}	Approved admins create/update; no delete.
events/{eventId}/documents/{documentId}	Event document references and external links. No Firebase Storage upload.	src/services/documentService.js	firestore.rules match /events/{eventId}/documents/{documentId}	Approved admins and event managers create/update; admins delete.
events/{eventId}/runOfShow/{itemId}	Event-day schedule sequence, dependencies, arrivals, status.	src/services/runOfShowService.js	firestore.rules match /events/{eventId}/runOfShow/{itemId}	Approved admins manage; assigned event staff can read through task read gate.
events/{eventId}/resources/{resourceId}	Equipment, supplies, packing, pickup, return, quantity tracking.	src/services/eventResourceService.js	firestore.rules match /events/{eventId}/resources/{resourceId}	Approved admins manage; assigned event staff can read through task read gate.
contacts/{contactId}	Reusable contact directory.	src/services/contactService.js	firestore.rules match /contacts/{contactId}	Approved admins create/update/read; delete false.
organizations/{organizationId}	Reusable organization directory.	src/services/contactService.js	firestore.rules match /organizations/{organizationId}	Approved admins create/update/read; delete false.
events/{eventId}/contactLinks/{linkId}	Event relationship links to contacts/organizations.	src/services/contactService.js	firestore.rules match /events/{eventId}/contactLinks/{linkId}	Approved admins and event managers create/update; admins delete.
accessRequests/{requestId}	Signed-in user access request workflow.	src/services/accessRequestContract.js	firestore.rules match /accessRequests/{requestId}	Signed-in create; approved admins review; no delete.
staffProfiles/{uid}	Global staff profile records.	src/services/staffManagementService.js; src/auth/AuthProvider.jsx	firestore.rules match /staffProfiles/{uid}	Approved admins manage; staff can read own active profile.

Indexes

Collection group	Fields	Purpose
---	---	---
registrations	eventId ASCENDING, createdAt DESCENDING	Supports scoped Firestore query ordering used by the app.

Schema Truth Rule

Do not infer schema from one example record. Cross-check service writes, page reads, utils, Firestore Rules validators, and tests.

<!-- Source: docs\manual\09-permissions-and-security-rules.md -->

Permissions and Security Rules

Permission architecture has two layers:

- 1. Frontend access checks in `src/utls/accessRoles.js` and `ProtectedRoute`.
- 2. Firestore Rules in `firestore.rules`.

```
flowchart TD
    Start[Request route or Firestore operation] --> Signed{Signed in?}
    Signed -- No --> Login[Redirect/login denied]
    Signed -- Yes --> Owner{Protected Owner UID?}
    Owner -- Yes --> Admin[Admin-level frontend access]
    Owner -- No --> Allowlist{Active approved organizer?}
    Allowlist -- Yes --> Admin
    Allowlist -- No --> Staff{Active staff profile + event assignment?}
    Staff -- No --> Deny[Deny access]
    Staff -- Yes --> Role{Assignment role}
    Role --> Scanner[Scanner: assigned event check-in only]
    Role --> Manager[Event manager/viewer: narrow event reads/tasks/docs]
    Role --> Ops[Operations helper: operations visibility only]
    Admin --> Rules[Firestore validates fields, audits, and event scope]
```

Permission Matrix

Action	Protected Owner	Approved Organizer/Admin	Assigned Staff	Scanner	Unapproved
Open organizer shell	Yes	Yes	Limited by assigned route	No, scanner uses /scanner	No
Manage Settings organizer access	Yes, immutable owner cannot be removed	No owner-only controls	No	No	No
Create/update events	Yes	Yes	No	No	No
Read assigned event	Yes	Yes	Yes if assignment active	Yes for scanner route	No
Create/update registrations	Yes	Yes	No	No except scanner check-in fields	No
Assign tickets	Yes	Yes	No	No	No
Complete check-in	Yes	Yes	No unless scanner assignment	Yes for assigned event	No
Use Operations ledger	Yes	Yes	operations-helper read only	No	No
Use Import Center	Yes	Yes	No	No	No
Read documents/tasks as assigned staff	Yes	Yes	event-manager/viewer where allowed	No	No
Create audit logs	Only with matching mutation	Only with matching mutation	Only where rules allow matching mutation	Only check-in audit path	No

Firestore Rules Reference

Important helper functions include `isSignedIn`, `isProtectedOwner`, `isApprovedAdmin`, `activeStaffProfile`, `activeStaffAssignment`, `isAssignedScanner`, `canReadTask`, and `canManageTask`.

Rules distinguish `resource.data` from `request.resource.data`, and create/read/update/delete paths are intentionally different. Query rules are not filters: if a query can return forbidden documents, Firestore denies the whole query.

<!-- Source: docs\manual\10-events-guests-tickets-and-checkin.md -->

Events, Guests, Tickets, and Check-In

Event Management

Events are managed by `EventsPage` and `eventService`. The Working Event context determines which event-scoped pages load data.

Guest Registrations

Registrations include guest identity, persons attending, payment state, ticket status, historical attendance evidence, and scanner-confirmed check-in fields. Registration mutations must stay atomic with audit evidence.

Tickets and QR

QR payload is exactly:

```
GSV:TICKET:{ticketCode}
```

Do not put private guest data into QR payloads.

Check-In Flow

```
flowchart TD
    Ticket[Ticket code] --> Payload[GSV:TICKET:{ticketCode}]
    Payload --> QR[QRCode rendering]
    QR --> Scanner[Camera or manual scanner panel]
    Scanner --> Parse[qrTicketUtils parses safe ticket code]
    Parse --> Lookup[Lookup registration in selected/assigned event]
    Lookup --> Confirm[Operator explicitly checks in]
    Confirm --> Firestore[Minimal checkedIn fields + audit log]
    Firestore --> Feedback[Success, duplicate, or permission feedback]
```

<!-- Source: docs\manual\11-imports-exports-and-qr-systems.md -->

Imports, Exports, and QR Systems

Import Flow

```
flowchart TD
    Source[CSV, XLSX, pasted table] --> Parse[Parser and source type]
    Parse --> Mapping[Header mapping and sheet confirmation]
    Mapping --> Preview[Preview rows: valid, warning, needs review, blocked]
    Preview --> Duplicate[Duplicate and ticket code checks]
    Duplicate --> Confirm[Explicit final confirmation]
    Confirm --> Batch[Chunked Firestore writes + audit logs]
    Batch --> Review[Registration records and follow-up]
```

Supported sources: CSV, XLSX through `read-excel-file`, and pasted table rows. The app uses preview-first validation, explicit sheet confirmation, duplicate detection, ticket-code collision checks, and chunked writes with audits.

Exports are client-side generated files from `src/utils/exportUtils.js` and related finance/reconciliation helpers. Audit log and access-control collections are not included in ordinary organizer exports.

<!-- Source: docs\manual\12-testing-and-quality-assurance.md -->

Testing and Quality Assurance

Command Reference

Command	Purpose	Actual script	When to run	Expected result
---	---	---	---	---
npm run dev	Local validation/development command	vite	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run build	Local validation/development command	vite build	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run preview	Local validation/development command	vite preview	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run docs:generate	Local validation/development command	node scripts/docs/generateTechnicalManual.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run docs:validate	Local validation/development command	node scripts/docs/validateTechnicalDocs.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run lint	Local validation/development command	eslint src tests scripts e2e eslint.config.js vite.config.js playwright.config.js	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run test	Local validation/development command	node --test tests/*.test.js	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run doctor	Local validation/development command	react-doctor . --project . --scope full --blocking none --no-telemetry	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run doctor:changed	Local validation/development command	react-doctor . --project . --scope changed --base main --blocking error --no-telemetry	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run doctor:json	Local validation/development command	react-doctor . --project . --scope full --blocking none --json --json-compact --no-telemetry	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:copy-scan	Local validation/development command	node scripts/product/copyScan.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:routes	Local validation/development command	node scripts/product/routeInventory.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:bundle	Local validation/development command	node scripts/product/bundleSummary.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:docs	Local validation/development command	node scripts/product/documentationCheck.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:legacy	Local validation/development command	node scripts/product/legacyControlCheck.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:qa	Local validation/development command	node scripts/product/runProductCommand.mjs qa	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:audit	Local validation/development command	node scripts/product/runProductCommand.mjs audit	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run product:qa:emulator-checks	Local validation/development command	node --test tests/firestore-checkin-rules.test.js tests/task-workflow-rules.test.js tests/document-contact-rules.test.js tests/run-of-show-resource-rules.test.js && npm run e2e:smoke:inner	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:smoke	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "npm run e2e:smoke:inner"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:smoke:inner	Local validation/development command	playwright test --project=chromium e2e/navigation.spec.js	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:full	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "playwright test --project=chromium"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:firefox	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "playwright test --project=firefox"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run e2e:webkit	Local validation/development command	npx -y firebase-tools@14.19.0 emulators:exec --only auth,firestore --project gathervibeshub "playwright test --project=webkit"	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:ensure-access	Local validation/development command	node scripts/admin/ensureAccessControl.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message
npm run admin:verify-firebase	Local validation/development command	node scripts/admin/verifyFirebaseSetup.mjs	During development, QA, or documentation validation	Exit code 0; command-specific pass message

```
| npm run admin:replace-codex-test-with-demo | Local validation/development command | node
scripts/admin/replaceCodexTestWithDemoEvent.mjs | During development, QA, or documentation validation | Exit code 0; command-
specific pass message |
| npm run admin:verify-production-fixtures | Local validation/development command | node
scripts/admin/verifyProductionFixtures.mjs | During development, QA, or documentation validation | Exit code 0; command-specific
pass message |
| npm run admin:verify-production-counts | Local validation/development command | node scripts/admin/verifyProductionCounts.mjs |
During development, QA, or documentation validation | Exit code 0; command-specific pass message |
| npm run firebase:deploy-rules | PRODUCTION-IMPACTING COMMAND where noted | npx firebase-tools deploy --only
firestore:rules,firestore:indexes --project gathervibeshub | Only after explicit approval and validation | Exit code 0; command-
specific pass message |
| npm run firebase:deploy-hosting | PRODUCTION-IMPACTING COMMAND where noted | npx firebase-tools deploy --only hosting --project
gathervibeshub | Only after explicit approval and validation | Exit code 0; command-specific pass message |
| npm run firebase:deploy-all | PRODUCTION-IMPACTING COMMAND where noted | npx firebase-tools deploy --only
firestore:rules,firestore:indexes,hosting --project gathervibeshub | Only after explicit approval and validation | Exit code 0;
command-specific pass message |
```

Testing Layers

- Unit and source contract tests: `tests/\*.test.js`.
- Firestore Rules tests: rules-specific test files run against emulators.
- E2E smoke: `e2e/navigation.spec.js`.
- Accessibility and responsive E2E: `e2e/accessibility.spec.js`, `e2e/responsive.spec.js`.
- Product QA wrapper: `npm run product:qa`.
- React Doctor: `npm run doctor:json` or `npm run doctor:changed`.

Date-Sensitive Test Policy

Phase 2 found `tests/document-contact-foundation.test.js` used `2026-08-20` as an expiring-soon fixture. On 2026-08-21 that fixture became expired for helpers using the current clock. It was updated to `2026-08-30` to preserve the intended behavior.

Future date-sensitive tests should either inject an explicit clock into every helper under test or choose dates far enough away to avoid silent expiry.

```
<!-- Source: docs\manual\13-build-deployment-and-recovery.md -->
```

Build, Deployment, and Recovery

Local Build

```
npm ci
npm run lint
npm test
npm run build
npm run product:qa
```

Firebase Project

Default project in `.firebaserc`: ``gathervibeshub``.
Hosting public folder: ``dist``.
Security headers and SPA rewrite are configured in ``firebase.json``.

Production-Impacting Commands

Command	Impact	Prerequisites
---	---	---
<code>npm run firebase:deploy-hosting</code>	PRODUCTION-IMPACTING COMMAND: deploys Hosting assets from dist.	Build passed; production visual check plan ready.
<code>npm run firebase:deploy-rules</code>	PRODUCTION-IMPACTING COMMAND: deploys Firestore Rules and indexes.	Rules tests passed; authorization impact reviewed.
<code>npm run firebase:deploy-all</code>	PRODUCTION-IMPACTING COMMAND: deploys Hosting, Rules, and indexes.	Use only when all targets intentionally changed.

Deployment Flow



No production deployment is performed by documentation generation.

<!-- Source: docs\manual\14-troubleshooting-and-repairs.md -->

Troubleshooting and Repairs

Use the runbooks in `docs/runbooks/` for step-by-step procedures.

Priority problems:

- Blank page or stale deployment chunk.
- Login or auth persistence failure.
- Missing or insufficient permissions.
- Owner account denied.
- Staff assignment missing.
- Scanner cannot check in.
- Firestore permits data that frontend hides or frontend shows data Firestore denies.
- Build, dependency, emulator, or deployment failure.
- QR scanner failure during an event.

<!-- Source: docs\manual\15-legacy-systems-and-technical-debt.md -->

Legacy Systems and Technical Debt

Current Register

Item	Classification	Notes
---	---	---
Archived migration evidence	archive only / future review required	Located at C:\Users\Jaylan\Documents\Archived Projects\gsv-codex-demo-migration-evidence-2026-08-21. Contains CPB/CODEX_DEMO evidence and old repo snapshots.
Legacy repository archive	archive only	Located at C:\Users\Jaylan\Documents\Archived Projects\gathervibes-legacy-repository-archive-2026-08-21.
21 node_modules in historical archive snapshots	future review required	Generated dependencies likely reclaimable, but recursive deletion was blocked in Phase 2.
Historical branches	future review required	Branches were not deleted in Phase 2. Review merge status before any cleanup.
Active copilot worktree	current / unknown future work	Retained at C:\Users\Jaylan\Documents\gathertr.worktrees\copilot-personalized-welcome-tour-implementation.
React Doctor warnings	technical debt	doctor:json reports warnings, but doctor:changed found no issues for Phase 2 changed file.
Legacy role aliases	legacy but required	accessRoles retains aliases such as checkInStaff for compatibility.

<!-- Source: docs\manual\16-change-management.md -->

Change Management

No meaningful coding task is complete until documentation impact has been reviewed.

Every handoff must report:

- Documentation reviewed: YES/NO
- Documentation changed: YES/NO
- Documentation files changed
- Reason no documentation update was required

Use templates in `docs/templates/` for change records, incidents, repair runbooks, release checklists, and manual update checklists.

<!-- Source: docs\manual\17-reference-and-glossary.md -->

Reference and Glossary

Glossary

- Approved Organizer: an active email in `settings/accessControl`.
- Protected Owner: the permanent UID-protected owner account, independent of mutable allowlists.
- Working Event: selected event context used by event-scoped pages.
- CODEX_DEMO: synthetic training event safe for reversible QA.
- CPB: real completed event; not a synthetic QA target.
- Audit log: append-only evidence document required for many business writes.
- Query rules are not filters: Firestore denies a query if it could return unauthorized documents.

Search Keywords

```
Firestore missing or insufficient permissions resource.data request.resource.data
Firestore query rules are not filters
React Firebase auth state loading before Firestore query
Vite dynamic import Failed to fetch dynamically imported module
Firestore Hosting stale service worker chunk cache
Playwright Firestore emulator auth firestore port taken 9099 8080
html5-qrcode camera permission HTTPS mobile browser
GSV:TICKET QR payload ticketCode privacy
read-excel-file browser XLSX import mapping
React Doctor no-loading-flag-reset-outside-finally
```

<!-- Source: docs\manual\18-future-app-development-reference.md -->

Future App Development Reference

Lessons from Gathetr that apply to future React/Firebase apps:

- Design the permission model before building pages.
- Keep frontend route gates and backend Rules aligned but separate.
- Use one authoritative source for access control and display it in Settings.
- Avoid synthetic QA against real event data.
- Keep QR payloads privacy-safe and stable.
- Add emulator-backed rule tests before deploying Rules.
- Document every schema, role, status, and workflow change at the time it changes.
- Keep generated PDFs as snapshots, not the source of truth.